Romanian Memes

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Abstract: The aim of this project is to check how precise we can classify text through natural language processing techniques. For this purpose, we make an experimental evaluation, given an input of Romanian memes, which will be classified by certain algorithms based on theme, feelings and political statements. The images provided may need certain corrections done by our programme prior to classification, such as resizing, binarization or denoise, so that the text is clear enough. After this phase, the project development follows a sequence of three steps which are applied in this exact order - brute OCR, normalization and adding diacritics. The resulting technology is a multimodal pipeline, which will make connections between the image itself and the text written on it. The final outcome is a comparison between the results of each step and the original dataset (as matching percentage), as well as between each other in order to check for progress.

Keywords: OCR, diacritics, Romanian, classification, text, image

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1. **Understanding the requirements**

1.1. *Briefly describe what you understood from the given assignment:*

We downloaded ~450 pictures with memes(RoMEMES dataset from Zenodo record) and extracted the text from each one of them. We take every image, apply different strategies to make the text more clear. With the new improved images, we use Tesseract/EasyOCR to extract the text from them. We apply diverse modifications to this text and compare the output with the real text from the dataset. Now, if we have the text from each image, we can start training an AI algorithm.

1.2. *What are the main objectives you need to achieve?*

We need to process the given images with certain libraries and check whether or not the algorithm classified them correctly, by comparing the results with the ones in the *metadata* file. We try to improve them by correcting the images and/or the text given by the OCR to achieve the best NLP.

2. **Implementation planning**

1.3. *What technologies or programming languages do you intend to use?*

Our core technology stack is built on Python for its extensive NLP ecosystem. We are utilizing PyTorch as our primary machine learning framework and the Tesseract OCR engine for high-performance optical character recognition

1.4. *What libraries or frameworks do you consider useful for your project?*

The libraries fit for the Romanian language analysis are:

- EasyOCR for flexible text extraction

- Teprolin for essential preprocessing like lemmatization
- RoWordNet for semantic mapping
- Romanian-Transformers for high-accuracy emotion detection.

1.5. *How will you organize the implementation stages?*

- **Phase 1:** Improve image quality
- **Phase 2:** Word extraction with OCR algorithm
- **Phase 3:** Text normalization
- **Phase 4:** Adding diacritics
- **Phase 5:** Classification (text-only and multimodal)
- **Phase 6:** Comparing the results and creating statistics (accuracy percentages, plotting graphs)

1.6. *What are the main technical challenges you anticipate?*

- Learning new technologies in order to use them properly
- Working with AI algorithms
- Analysing and working with big dataset

1.7. *How will you manage resources and allocate time for each phase?*

We will work on the first four phases asynchronously in order to manage time efficiently since this will be the most complex part of the project. After assembling the resulting implementations, we can move on to the classification algorithm, which will be a group effort because it involves all the previous code. For the last phase of our project, we will work in parallel once again, creating statistics on the performance of each of the first phases.

Each person will work on their personal laptop and upload the code on GitHub, in a shared repository. When working asynchronously, we will use branches and pull requests so that the code remains organized.

1.8. *How will you distribute responsibilities for each teammate? (please specify explicitly for each one).*

We are going to create 3 teams of 2 members in order to work simultaneously:

- Alexandra and Andrei are going to improve image quality and implement OCR algorithm
- Flavia and Marian are going to implement normalization on the extracted text (working on a sample to begin with)
- Maria and George are going to add diacritics to the normalised text (or to a sample text at first)

After the pipeline is done, we are going to implement the classification algorithms together and apply it to each of the 3 steps to see the results.